Predicting Customer Churn in the Telecommunication Industries Using Machine Learning Techniques

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Declaration and Approval

Declaration

I declare that this work has not been previously submitted and approved for the award of a degree by this or any other University. To the best of my knowledge and belief, the proposal contains no material previously published or written by another person except where due reference is made in the proposal itself.

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Abstract

Customer churn significantly impacts profitability and competitiveness in the telecommunication sector, making effective prediction and prevention strategies essential. This proposal outlines a study leveraging machine learning models to predict churn based on key customer behavior and service usage patterns collected from a telecommunication Industry. The research focuses on variables that may affect the churn rate. Multiple models are compared, with hyperparameter tuning applied to enhance predictive reliability. The best performing model is deployed within an interactive dashboard, enabling stakeholders to identify at risk customers quickly and take proactive retention measures. By integrating advanced analytics into decision-making, the approach supports operational efficiency, reduces acquisition costs, and fosters long-term digital engagement, especially in emerging markets.

Telecommunication companies face increasing competition and market saturation, making customer retention a critical driver of revenue stability. Predictive analytics offers the potential to transform churn management by enabling timely interventions based on behavioral insights. Traditional churn management relies on reactive approaches, often identifying at risk customers too late. Manual or rule-based methods are time-consuming, computationally inefficient, and fail to capture complex patterns in customer behavior.

We propose the development of a machine learning–driven churn prediction system using telecommunication data. Models including Logistic Regression, Decision Tree, Random Forest, and LightGBM will be evaluated using classification metrics. Hyperparameter tuning will be applied to optimize model performance. The final model will be deployed using Django framework, enabling real-time identification of at-risk customers and supporting targeted retention strategies.The proposed solution will provide a robust, data-driven tool for proactive churn management, enabling telecommunications providers to retain valuable customers. This approach will enhance decision-making efficiency and contribute to sustained competitiveness in the industry.

Keywords: Random Forest, Customer Churn, Telecommunication, Customer retention

Acronyms

1. ML - Machine Learning
2. DL - Deep Learning
3. LMT - Logistic Model Tree
4. RF - Random Forest
5. FT - Functional Trees
6. LR - Logistic Regression
7. DT - Decision Tree
8. k-NN - k-Nearest Neighbors
9. ERT - Extra Random Trees
10. GBM - Gradient Boosting
11. ANNs - Artificial Neural Networks
12. CNNs - Convolutional Neural Networks

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1. Introduction

1.1 Background

Customer churn defined as customers discontinuing their subscription or services in favor of competing offers represents a critical and persistent challenge within the global telecommunications industry. Churn profoundly affects business profitability, operational sustainability, and long-term market competitiveness. Globally, annual churn rates in telecommunications range between 20% and 40%, contributing to substantial revenue losses for service providers each year and underscoring the urgency for proactive prediction and retention strategies *Annual Churn Rate in Global Telecommunications Sector* (2023).

From an economic perspective, customer retention is significantly more cost-effective than acquisition. Studies indicate that acquiring a new customer can be five to seven times more expensive than retaining an existing one Kotler et al. (2021). High churn rates not only inflate acquisition costs but also erode market share, reduce revenue stability, and constrain organizational competitiveness. Socially and culturally, stable telecommunication access is essential for enabling critical services, supporting digital inclusion, and facilitating economic participation, particularly in emerging economies such as those in Sub-Saharan Africa *Mobile Economy Report: Sub-Saharan Africa 2022* (2022).

Despite ongoing investment in churn reduction strategies, many telecommunications companies still rely on traditional methods of churn analysis and prediction. These methods often use static and retrospective datasets that inadequately capture the dynamic shifts in customer preferences, external socio-economic influences, and seasonal or event-driven fluctuations in usage Jahromi and Sharifi (2020). Such approaches frequently lead to subjective decision-making, inefficient allocation of retention resources, and limited predictive accuracy.

Recent advances in data science, particularly in Machine Learning (ML), present an opportunity to address these limitations. Predictive analytics, leveraging algorithms such as Logistic Regression, Decision Trees, and Random Forest Classifiers, have demonstrated strong potential for modeling complex customer behaviors and identifying at-risk subscribers before they leave Rahman et al. (2024). These models can integrate large-scale, heterogeneous datasets, uncover non-linear patterns, and generate actionable insights with higher accuracy than conventional approaches.

In Kenya's telecommunications market is characterized by high competition, rapid digital adoption, and socio-economic reliance on mobile services, these analytical capabilities are particularly relevant. By combining behavioral, demographic, and service usage data with advanced ML techniques, providers can proactively intervene to retain customers, optimize marketing expenditure, and strengthen customer loyalty.

1.2 Research Problem

The Kenyan telecommunications sector faces intensifying competition, evolving consumer expectations, and increased vulnerability to churn due to competitive offers and service quality perceptions. Current churn prediction and management approaches in many organizations remain reactive, relying on static indicators and manual analyses that are time-consuming, computationally inefficient, and prone to human bias. These methods struggle to capture real-time changes in customer behavior and broader socio-economic shifts, resulting in sub-optimal retention strategies and high revenue leakage.

1.3 Research Objectives

This study seeks to address the identified gaps through the following objectives:

- To identify key churn predictors in telecommunication industry.
- To develop a machine learning based churn prediction model using Kenyan telecommunications data.
- To evaluate and compare the performance of multiple ML algorithms.
- To design and deploy an interactive dashboard integrated into telecommunication providers' CRM systems for real-time churn risk monitoring and intervention.

1.4 Significance and Justification

From a global perspective, effective churn management is critical for sustaining profitability and competitiveness in the telecommunications sector. Retention-focused strategies not only reduce the high costs associated with customer acquisition but also stabilize revenue streams, safeguard market share, and enhance long-term customer value.

In emerging markets like Kenya, the implications of churn extend beyond corporate profitability. Telecommunications services are deeply embedded in daily life, acting as enablers for financial transactions, e-learning, e-health, and small business operations. High churn rates risk excluding segments of the population from these benefits, slowing progress toward digital inclusion and socio-economic development.

This research is justified on three key fronts:

Economic efficiency: By identifying at-risk customers before they leave, providers can target retention efforts more precisely, thereby reducing costs and maximizing return on marketing investment.

Operational effectiveness: The integration of the predictive model into CRM systems ensures actionable insights are delivered to decision-makers in real time, enabling timely and effective interventions.

Scalability and replicability: While tailored to Kenya's socio-economic and competitive environment, the proposed framework offers a methodological blueprint that can be adapted for other markets and industries facing similar churn-related challenges.

Ultimately, this study contributes to both the academic body of knowledge on churn prediction and the practical toolkit available to telecommunications providers. By leveraging advanced ML techniques and embedding them into operational workflows, the research aims to create a sustainable, data-driven approach to customer retention in highly competitive markets.

2. Literature Review

The telecommunications industry has undergone substantial transformation over the past decade, driven by technological advancements, market liberalization, and evolving consumer expectations. While these developments have expanded access and improved service delivery, they have also intensified competition among service providers, making customer retention a critical strategic priority. One of the most pressing challenges in this context is customer churn the phenomenon where subscribers discontinue their relationship with a service provider in favor of competitors.

2.1 Theoretical Framework

Customer churn, as defined by Sharma and Panigrahi (2011), refers to a customer terminating their relationship with one company to join a competitor. Churn directly reduces revenue streams and has broader operational impacts, including increased marketing expenditure to replace lost customers Chen et al. (2014). Hadden et al. (2005) describe churn management as the systematic identification of customers likely to defect, while Risselada et al. (2010) frame it as an integral component of customer relationship management (CRM). Effective churn management seeks to extend customer lifetime value and safeguard market share in competitive environments.

2.1.1 Customer Churn in Service Industries

In service industries, particularly those with subscription based models, customer loyalty is influenced by multiple factors, including perceived service quality, pricing structures, and customer experience. Smith et al. (2021) demonstrated that integrating quantitative usage metrics with qualitative customer feedback and sentiment analysis offers deeper insights into the drivers of churn. Their findings underscore the need for telecommunications providers to adopt holistic churn management approaches that address both operational performance and customer perception.

2.1.2 Technology Adoption Theory

Rogers' (2003) Diffusion of Innovations theory provides a relevant lens for understanding how customers adopt, continue, or discontinue the use of telecommunications services. Adoption decisions are shaped by perceived relative advantage, compatibility with user needs, and complexity of use. Within the churn context, slow adoption of new service offerings or dissatisfaction with technological updates can influence a subscriber's decision to switch providers. Applying this theory helps to contextualize behavioral patterns in churn prediction models.

2.1.3 Relationship Management (CRM)

Payne and Frow (2005) highlight the strategic role of building and maintaining customer relationships to enhance retention. In telecommunications, CRM strategies such as personalized offers, targeted engagement, and loyalty programs can reduce churn risk. Embedding CRM principles into churn prediction initiatives strengthens their practical applicability by aligning predictive insights with actionable retention strategies.

2.2 Empirical Studies

Tuzovic and Kabadayi (2018) identify key churn drivers in mobile-based services, including poor service quality, lack of personalization, and competitive offers. In the telecommunications context, similar factors manifest through dropped calls, network downtime, rigid pricing models, and inadequate customer service. Understanding these drivers is foundational for developing predictive models that accurately reflect real-world churn determinants.

Recent advances in machine learning and big data analytics have enabled more accurate churn prediction. Usman-Hamza et al. (2022) demonstrated that ensemble decision forest models, including Logistic Model Trees (LMT), Random Forest (RF), and Functional Trees (FT), enhanced with weighted soft-voting stacking, outperform baseline classifiers particularly in class imbalanced datasets common in telecommunication churn.

Comparative studies have shown that deep learning models, especially Convolutional Neural Networks (CNNs), can achieve optimal trade-offs between accuracy and profitability when compared with traditional models such as Logistic Regression (LR) and Decision Trees (DT). Saha et al. (2023) reported that CNNs performed exceptionally well in hybrid datasets from India and the U.S., highlighting the value of ROI-sensitive model selection for telecommunication providers.

Nagarkar (2022) applied XGBoost to a large Nepalese telecommunications dataset, leveraging behavioral predictors such as recharge frequency and inactivity duration. Although specific metrics were not disclosed, the model informed targeted seasonal promotions and reduced revenue loss. These results demonstrate the potential of tailoring models to the socio-economic and operational context of the market.

Ahmad et al. (2019) integrated SNA-derived features into churn models for SyriaTel using Decision Trees, RF, GBM, and XGBoost. Processed via Apache Spark, the inclusion of SNA features improved AUC from 0.84 to 0.933, illustrating the predictive value of relational data in understanding churn behavior.

Phua et al. (2012) demonstrated the value of temporal utilization indicators and payment history in both churn and win-back predictions. By applying Random Forest and SimpleCart on imbalance corrected datasets, they revealed that longitudinal usage patterns are strong predictors of customer retention or defection.

Ahmed et al. (2019) transformed structured telecommunication datasets into image based representations for CNN processing, integrating GP AdaBoost stacking. This transfer learning approach achieved accuracies of 75.4% and 68.2% with AUC scores of 0.83 and 0.74, highlighting opportunities for cross-domain methodological innovation.

Shaikhsurab and Magadum (2024) proposed an adaptive ensemble incorporating XGBoost, LightGBM, LSTM, MLP, and SVM, stacked with meta-features, achieving 99.28% accuracy on public telecom datasets. While performance was exceptional, interpretability challenges remain a barrier to operational adoption.

Emerging literature emphasizes evaluating models not only on statistical accuracy but also on profitability. Profit-sensitive evaluation frameworks align churn prediction outputs with the financial objectives of the business, ensuring that retention interventions maximize return on investment.

2.3 Integration of Theoretical and Empirical Insights

The theoretical frameworks reviewed highlight the economic, behavioral, and relational dimensions of churn, while empirical studies showcase the power of advanced analytics in predicting and mitigating churn. Integrating CRM principles, technology adoption theory, and economic perspectives with machine learning methods provides a comprehensive foundation for developing effective churn management solutions in the telecommunications sector.

2.4 Customer Churn in the Telecommunications Industry

Customer churn in telecommunications poses a substantial threat to revenue stability and market position. High churn rates increase customer acquisition costs and reduce profitability. Research by Khan et al. (2024) confirms that annual churn rates in global telecommunication operations can exceed 30%, underscoring the urgency for robust predictive and retention strategies.

2.5 Industry Relevance and Contribution

Predicting and reducing churn is critical to the sustainability and growth of telecommunications providers. While numerous predictive models exist, there is limited research applying data science methodologies specifically to African telecommunication markets. This study addresses that gap by focusing on the Kenyan telecommunications sector, leveraging advanced machine learning techniques and operational integration via CRM systems.

2.6 Gaps in Existing Literature

- Underutilization of Unstructured Data: Text logs, social media sentiment, and other contextual sources remain largely untapped.
- Geographic Gaps: Few studies address African telecommunications markets, especially in rural and low-resource contexts.

2.7 Conceptual Framework

The conceptual framework for this study comprises the following components:

- User Behavior Analysis: Examination of customer demographics, usage patterns, contractual terms, and payment behaviors to identify churn predictors.
- Machine Learning Models: Implementation of algorithms including Logistic Regression, Random Forest, Decision Trees, and Gradient Boosting to predict churn, with comparative performance analysis.
- User Retention Strategies: Translation of model outputs into actionable retention initiatives tailored to the Kenyan telecommunications market.
- Performance Evaluation: Use of metrics such as accuracy, precision, recall, F1-score, and ROC-AUC to assess predictive performance.
- Contextual Considerations: Incorporation of Kenya's socio-economic and operational realities to ensure recommendations are practical and market-relevant.

3. Methodology

This study adopts the Cross-Industry Standard Process for Data Mining (CRISP-DM) methodology to ensure a structured, repeatable, and scalable approach to customer churn prediction in the Kenyan telecommunications sector.

3.1 Data Understanding

We intend to collect data through a structured questionnaire administered to staff across various departments of a telecommunications operator. The questionnaire will be designed to capture detailed insights into customer demographics (e.g., gender, segment), contractual arrangements (e.g., contract type, tenure), financial aspects (e.g., typical monthly charges, total billed amounts), and service usage patterns (e.g., average data consumption, prevalent phone service types). The questions will focus on staff observations and departmental records of customer interactions, enabling the capture of both quantitative indicators and qualitative perspectives on customer behavior and churn. Personal identifiers will not be collected, and all data handling will adhere to relevant governance and confidentiality guidelines as per *Regulation (EU) 2016/679 of the European Parliament and of the Council* (2018).

A preliminary exploratory data analysis (EDA) will be conducted on the compiled questionnaire responses to categorize churn-related and non-churn-related patterns, identify potential class imbalances, and examine variable relationships. Statistical summaries, cross-tabulations, and visualizations such as correlation matrices and heat maps will be used to detect associations between billing-related variables and independent behavioral indicators reported by staff.

3.2 Data Collection

Data will be sourced through a structured questionnaire administered to staff across various departments of a Kenyan telecommunications operator. The questionnaire will be designed to capture insights on customer demographics, contractual arrangements, service usage patterns, billing processes, and perceived drivers of churn based on staff interactions with customers. Responses will also include departmental perspectives on customer behaviors, retention challenges, and service quality trends. Each participant will be assigned a unique anonymized identifier to ensure confidentiality and compliance with data governance guidelines. This approach will provide a multi-dimensional dataset incorporating both operational knowledge from staff and observed patterns of customer engagement, enabling a richer understanding of churn dynamics.

3.3 Data Cleaning

Data cleaning will constitute a critical component of the data pre-processing phase, undertaken to enhance the quality, accuracy, and reliability of the dataset prior to analysis. Given that the data will be obtained through structured questionnaires, potential sources of error may include typographical mistakes, incomplete responses, inaccurately recorded values, and inconsistencies in categorical labels or formatting.

A systematic cleaning process will be applied to address these issues. Specifically, missing values will be imputed using appropriate statistical or rule-based methods to preserve the representativeness and distributional characteristics of the data. Erroneous entries and illogical values will be identified and corrected through consistency checks and cross-validation against predefined logical rules. Furthermore, categorical and numerical variables will be standardized to ensure uniformity across all responses, thereby facilitating compatibility with subsequent analytical procedures.

This rigorous data cleaning protocol is intended to minimize noise, reduce the risk of bias, and ensure that the final dataset is robust, coherent, and fit for reliable statistical analysis and predictive modelling.

3.4 Machine Learning Modelling

3.4.1 Data Exploration

Through data exploration, this study will derive valuable insights into the distribution, variability, and structure of the collected dataset. This phase will involve performing descriptive statistical analyses, generating visualizations, and computing summary statistics to identify patterns, trends, and potential relationships within the data. Exploratory Data Analysis (EDA) techniques, including histograms, scatter plots, box plots, and correlation matrices, will be used to assess the relationships between customer demographic factors, contractual attributes, financial metrics, and service usage patterns. These insights will inform the selection of appropriate machine learning models and preprocessing strategies.

3.4.2 Treating Missing Data

Addressing missing data is critical to ensuring the completeness and quality of the dataset. Missing values will be handled based on their nature and extent. Where feasible, imputation techniques such as mean imputation, regression-based imputation, or domain-informed replacement will be applied. Where the proportion of missing data is excessive or the missingness is deemed non-random, the affected entries may be excluded to prevent bias. The choice of imputation method will be guided by the variable type, data distribution, and potential impact on predictive performance.

3.4.3 Treating Outliers

Outliers have the potential to distort statistical summaries and reduce the accuracy of predictive models. Outlier detection will be conducted using statistical thresholds (e.g., z-scores, interquartile ranges) and visual diagnostics (e.g., box plots, scatter plots). Detected outliers will be evaluated in consultation with subject matter experts to determine whether they represent valid extreme cases or erroneous entries. Appropriate treatments, such as capping, transformation, or removal, will be applied as necessary.

3.4.4 Data Type Conversion

Data type conversion will ensure that variables are consistently represented for analysis. Categorical variables will be converted into standardized label formats, and numerical variables will be cast to appropriate numeric data types. This process ensures compatibility with machine learning algorithms and facilitates reproducibility.

3.4.5 Data Transformation

Feature Engineering

The dataset will be refined to improve model predictive capacity through feature engineering. This process will involve generating new variables or modifying existing ones to capture latent patterns in customer churn behavior. Examples may include the computation of customer tenure categories, normalized spending ratios, and derived usage intensity scores.

Feature Scaling

Feature scaling will standardize the range of numerical variables to ensure equitable influence on model training. Standardization using z-score normalization and normalization using Min-Max scaling will be considered. These methods prevent variables with large numerical ranges (e.g., monthly charges) from dominating the learning process.

Feature Selection

Relevant features will be selected to reduce dimensionality, mitigate overfitting, and improve interpretability. Statistical tests, correlation analysis, and model-based feature importance rankings will guide the selection process. Irrelevant or redundant features will be excluded to enhance computational efficiency.

Data Encoding

Categorical attributes will be transformed into numerical representations suitable for machine learning models. One-hot encoding, label encoding, or binary encoding will be applied depending on the nature of the variable and the target algorithm's requirements.

3.4.6 Model Selection

Customer churn prediction constitutes a binary classification task, requiring the accurate categorization of customers into churn and non-churn groups based on their demographic, contractual, financial, and behavioral attributes. To address this challenge, the study will adopt a comparative modelling approach that evaluates several machine learning algorithms, each chosen for its unique strengths and suitability for the churn prediction problem in the telecommunications sector.

Logistic Regression will be employed as a foundational benchmark model. Its simplicity, computational efficiency, and ease of interpretation make it a reliable baseline for classification tasks. By estimating churn probabilities using a linear combination of input features, LR provides a clear understanding of how each predictor influences churn likelihood, which is valuable for decision-making and communication with non-technical stakeholders.

Random Forest, an ensemble method that constructs multiple decision trees and aggregates their predictions, will be used for its ability to handle large, heterogeneous datasets while maintaining high predictive performance. RF reduces the risk of overfitting, provides inherent measures of feature importance, and is robust to noisy data—qualities that are crucial when working with real-world telecommunications datasets.

Decision Tree models will be included for their interpretability and straightforward visualization of decision rules. DTs recursively split the dataset into smaller, more homogeneous groups based on feature values, allowing for intuitive mapping of customer attributes to churn outcomes. This transparency supports stakeholder trust and facilitates actionable insight generation.

Light Gradient Boosting Machine will be utilized for its state of the art performance and computational efficiency. LightGBM is optimized for large scale and high dimensional data, making it highly suitable for churn prediction in contexts where the number of features and records is substantial. Its gradient boosting framework enables the capture of complex, non-linear relationships between customer characteristics and churn behavior.

An ensemble model will be developed to combine the strengths of multiple base learners, enhancing robustness and stability in predictions. By aggregating the outputs of diverse algorithms, the ensemble approach is expected to reduce individual model biases and variances, thereby improving overall generalization performance.

Finally, a Logistic Regression model trained on data augmented using the Synthetic Minority Oversampling Technique (SMOTE) will be implemented to specifically address class imbalance. Churn datasets often contain disproportionately fewer churners compared to non-churners, and SMOTE will synthetically generate new minority class samples within the training set. This approach is expected to improve the model's sensitivity in detecting churners, which is essential for proactive retention strategies.

Through this multi-model strategy, the study will ensure a rigorous evaluation of algorithmic performance, leading to the selection of a final model that optimally balances predictive accuracy, interpretability, and operational applicability within the telecommunications context.

3.4.7 Model Evaluation

The models will be trained using an 80/20 train-test split with stratified sampling to preserve churn class proportions. Performance will be evaluated using:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}, \quad \text{F1-Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

The Receiver Operating Characteristic (ROC) curve and the Area Under the Curve (AUC) will be computed to measure the model's discriminatory ability. Given the business cost of false negatives in churn prediction, recall will be prioritized during model ranking.

3.5 Model Deployment

The selected model will be deployed via an Application Programming Interface (API) to enable integration with existing customer management systems. Deployment through an API will provide modularity, maintainability, and scalability, allowing the predictive engine to serve multiple client applications concurrently.

Deployment Framework: FastAPI will be used as the deployment framework due to its speed, built-in data validation, automatic documentation, and asynchronous capabilities. The deployed model will return churn probability scores and key predictive factors for each customer record. These outputs will support targeted retention campaigns and ongoing monitoring of churn risk.

4. Expected Outcomes

The proposed machine learning–driven churn prediction system is expected to deliver a robust, data-driven decision-support framework for the Kenyan telecommunications sector. By systematically integrating customer demographics, contractual information, financial metrics, and behavioral usage patterns, the solution will provide actionable insights for retention teams.

From a technical standpoint, the final deployment will feature:

High predictive performance across recall, precision, F1-score, and ROC-AUC, ensuring that at-risk customers are identified accurately and in time to act. An interpretable model output through Shapley value explanations, enabling business stakeholders to understand key churn drivers and design targeted retention campaigns. A scalable architecture integrated into the operator's CRM via a Django-based dashboard, capable of processing high-volume customer data without latency.

From an operational perspective, the outcomes will enable:

Proactive customer retention strategies, reducing revenue loss by identifying high risk customers before service termination. Optimized marketing and resource allocation, focusing retention efforts on customers with the highest risk and business value. Continuous monitoring and model adaptation, with automated AUC recalculation and feature-importance tracking to maintain model relevance as customer behaviors and market conditions evolve.

Ultimately, this research will yield a deployable, real-world solution that addresses the immediate business need for churn reduction and also contributes a replicable methodological framework applicable to other African telecommunications markets and related service industries.

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